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STD: CSE SEC: ..... SUBJECT: Computer Networks

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Ex:No:1
2017/26.

1. Various Network commands in Linux & Windows.

Aim:

To study of various network commands used in Linux & windows.

Basic NetworKing commands:

1) arp -a : (terminal) ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

Output:

gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3.

2) hostname : (terminal) This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output:

fedora.

3) ipconfig/all : (command)

This command displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter in your system.

Output: Hostname: laptop-MHUMQBR6.
Physical address: 0A-00-27-00-00-0A.
DHCP enabled : No
Subnet mask : 255.255.255.0
IPV4 address : 192.168.56.1

(Command).

4) nbtstat -a : This command help solve problems with NetBios name resolution.

Output:

-a : Lists the remote machine's name table given its name.

-n : List local NetBios names;

-r : List names resolved by broadcast and via WINS.

5) netstat : netstat displays a variety of statistics about a computers active TCP/IP Connections.
(Command)

Output:

Proto	Local address	foreign address	state
TCP	10.106.210.192:60268	a23-55-47-124: http.	time_wait.
TCP	10.106.210.192:63441	202:4070	Established.
TCP	127.0.0.1:7768	Laptop-MHUMQBR6: 52692	Established.

6) nslookup : (name server lookup) is a tool used to perform DNS lookups in Linux.
(Command)

Eg: -nslookup: www.google.com.

Output:

Default Server : Unknown.

Address : 10.106.210.238.

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7) pathping : Pathping is unique to window's and is basically a combination of ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output:

Server : 10.106.210.238.

Address : 10.106.210.238#53.

8) ping : (Packet INternet groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to each other devices.

Output:

64 bytes from fedora (fe 80::96 ff:454d:ce50) : icmp-seq =
ttl=64 time = 0.088 ms.

PING 24.21.4.8 (24.21.4.8) 56(84) bytes of data.

9) Route : Command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Output:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use
default	-gateway	0.0.0.0	UG	100	0	0
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0	0

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Some important Linux networking commands:

1) ip.

* Every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems.

Eg: ip address shown

link (lookup 00:00:00:00:00:00 and
00:00:00:00:00:00)
inet (127.0.0.1.18' scope host lo).

2) if config:

It is a staple in many sysadmin's tool belt for configuring and troubleshooting networks, It has since been replaced by the ip command discussed above.

3) mtr:

* Serves as a network diagnostic and trouble shooting tool.

Eg: mtr google.com

Packets:

loss	snt	last	avg	best	Wst
0.0%	178	0.1	0.1	0.0	0.1

4) tcp dump.

Designed for capturing and displaying packets.

Eg: tcpdump -D

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5) ping.

* It is a tool that verifies IP-level connectivity to another TCP/IP computers by sending ICMP echo request messages.

Eg: ping google.com,

Student Observation:

1) Which command is used to find the reachability of a host machine from your device?

⇒ ping command is used to find the reachability of a host machine from your device.

2) Which command will we give the details of hops taken by a packet to reach its destination?

⇒ mtr, it shows route from a computer to a specified host mtr provides a lot of statics about each hop such as response time and percentage.

3) Which commands displays the ip configuration of your machine.

⇒ ip command can show address information manipulate routing plus display network various devices interfaces & funnels.

4) Which command displays the TCP port states in your machine?

⇒ net stat.

It displays variety of statistic about a computer active TCP/IP connection.

5) Write the modify the ip configuration in a Linux machine.

⇒ * Assigning IP address to interfaces ip address add 192.168.1.254/24.

* Deleting IP address.

ip address del 192.168.1.254/24. device enp-S03.

Ex:No:2

Ex:2 - Study of Network cables.

Aim:

To study of diffrent types of Network cables and make a cross-wired cable and straight through cable using clamping/crimping tool.

a) Understand diffrent types of Network cable:

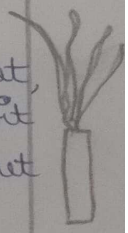
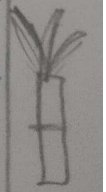
Diffrent type of cables used in networking are

i) Unshielded twisted pair (UTP) cable,

ii) Shielded twisted Pair (STP) cable,

iii) Coaxial cable,

iv) Fibre optic cable.

Cable type	Category	Maximum data transmission	Advantages/ Disadvantages:	Use	Image
UTP	Category 3	10 bps	Advantages: * cheaper in cost. * Easy to install Disadvantages: * More prone to (EMI).	10 base-T ethernet	
	Category 5	Up to 100 mbps		fast ethernet, gigabit ethernet	
	category 5e	1Gbps			
STP	Category 6,6a	10 Gbps	Advantages: * Shielded. * Faster than UTP. * Less susceptible to noise and interference.	gigabit, ethernet, 10G ethernet (55m) widely used in data centres.	

SSTP	Category 7	10 Gbps	Disadvantages: * Expensive * Greater installation effort.	gigabit ethernet, 10G ethernet.
Coaxial cable	RG-6. RG-59 RG-11	10-100Mbps.	* High bandwidth * Low loss Bandwidth Disadvantages: * Limited distance. * Cost. * Size is bulky.	Speed of signal is 500m television network high speed internet Connections.
fibre optics cable.	Single mode Multi mode	100 Gbps	Advantages: * High speed * High Bandwidth * High Security. Disadvantages: * Expensive. * Requires skilled installers.	* Maximum distance of fibre optics cable is around 100 meters.

b) Make your own ethernet Cross - Over cable / Straight cable.

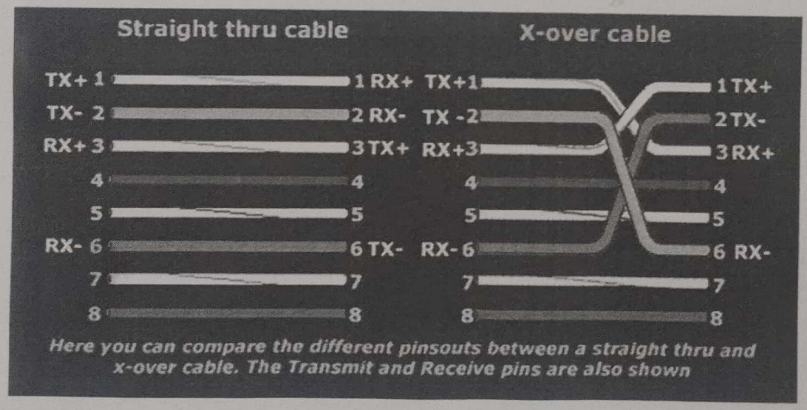
Tools and parts needed;

* ethernet cabling. CAT 5e is certified for gigabit support, but CAT 5 cabling works as well just over shorter distances.

* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and crimp and cut the shielding off the cables.

* Two RJ45 plugs,

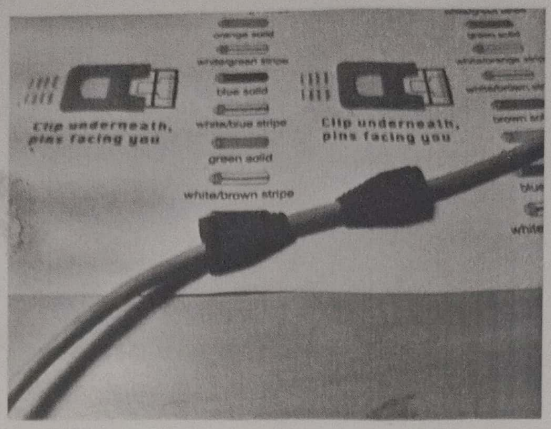
* Optional two plug shields.



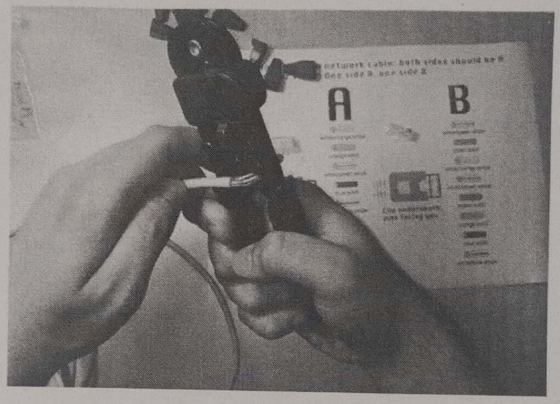
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

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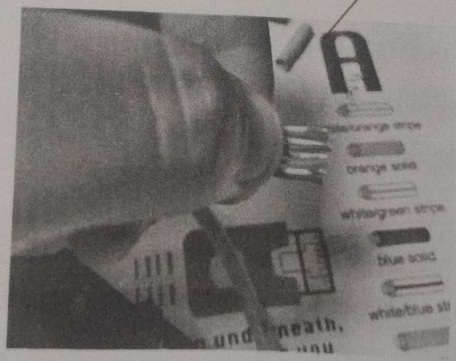
Step 1: To start construction of the device, begin by threading shields onto the cable,



Step 2: Next, strip approximately 1.5 of cable shielding from both ends. The crimping tool has a round area to complete this task.

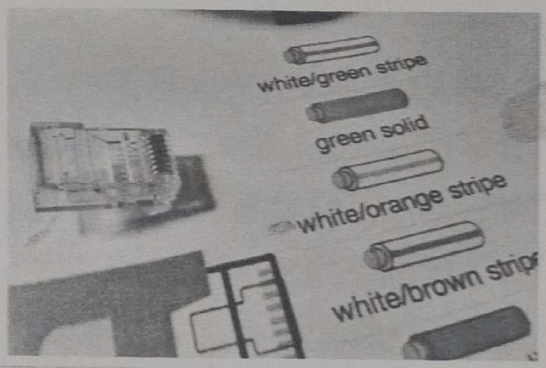


Step 3: After, you will need to untangle the wires; there should be four "twisted pairs", referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

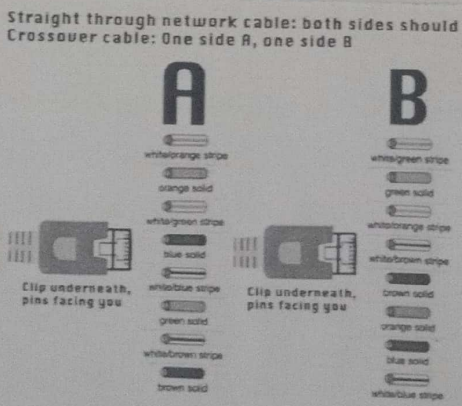


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Step 4:
Once the order is correct, bunch them together in a line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown



Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped off too much shielding. Simply snip the cables back a little more.



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Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B using diagram A.

To test it, plug it in and attempt to connect two devices directly.

Student observation:

1) What is the difference between Cross cable and straight cable?

⇒

	Straight cable	Cross cable.
*	Same wiring standard on both ends.	* Different wiring standards on each end.
*	Used to connect different types of devices.	* Used to connect similar types of devices.
*	Eg: PC to switch, Router to Switch.	* Eg: PC to PC, switch to switch.

2) which type of cable is used to connect two PCs?

⇒ Cross cable because both PCs are similar devices, so the transmit (TX) and receive (RX) wires must be crossed to communicate directly.

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3) which type of cable is used to connect a router/ switch to your PC?

⇒ Straight cable, because a PC and a switch/ router are different types of networking devices.

4) Find out the category of twisted pair cable used in your lab to connect the PC to the network socket.

⇒ Category: Cat5e or Cat6.

5) write down your understanding, challenges faced, and output received while making a twisted pair cross / straight cable.

⇒

* I learned the difference between straight and cross cables.

* I understood the T568A and T568B wiring standards.

* Arranging the wire colors in the correct order.

* Successfully prepared a straight / cross twisted pair cable.

* Tested the cable using a cable tester.

Ex: No: 3 Ex: 3 - Experiments on CISCO Packet Tracer (simulation tool).

Aim:

To study the packet tracer tool installation and user interface Overview.

c) To understand environment of CISCO Packet tracer to design simple network.

Introduction:

1) It allows you to model complex systems without the need for dedicated equipment.

2) It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.

3) It is available for both the linux and windows desktop environments.

4) Protocols in Packet tracker are coded to work and behave in the same way as they would on real hardware.

Installing Packet tracer.

To download Packet tracer, go to https://www.netacad.com and log in with your cisco networking academy credentials; then, click on the packet tracer graphic and download the package appropriate for your operating system.

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User interface overview:

The layout of packet tracer is divided into several components. The components of the packet tracer interface are as follows; match the numbering with explanations.

1) Menu bar - This is a common menu found in all software applications; it is used to open, save, print, change preferences, and so on.

2) Main toolbar - This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo and redo and on the right-hand side is an icon for entering network information for the current network.

3) Logical/physical workspace tab - These tabs allow you to toggle between the logical and physical work areas.

4) Workspace - This is the area where topologies are created and simulations are displayed.

5) Common tools bar - This toolbar provides controls for manipulating topologies.

6) Real time / Simulation tabs - These tabs are used to toggle between the real and simulation modes.

7) Network component box - This component contains all of the network and end devices available with packet tracer and is further divided into two areas; Device type selection box and Device-specific selection box.

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c) User-created packet box - users can create highly customized packets to test their topology from this area, and the results are displayed as a list.

d) Analyse the behaviour of network devices using Cisco packet tracer simulator:

i) From the network component box, click and drag and drop the below components.

a) 4 generic PCs and One hub.

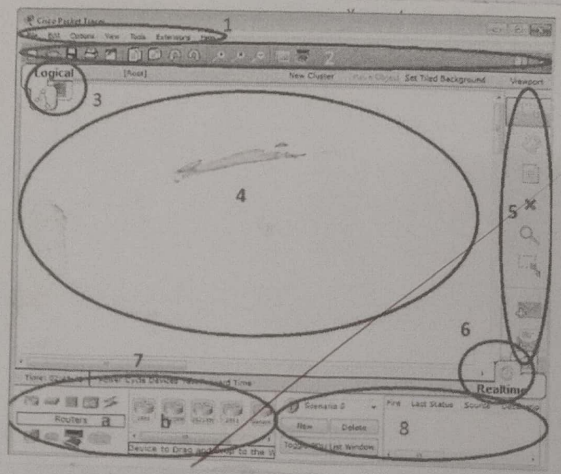
b) 4 generic PCs and one switch.

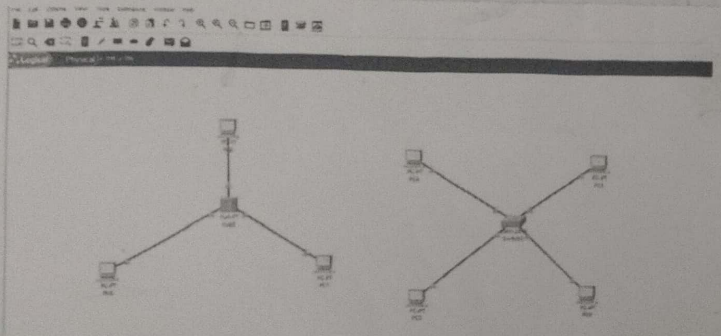
ii) Click on connections,

a) Click on copper straight-through cable.

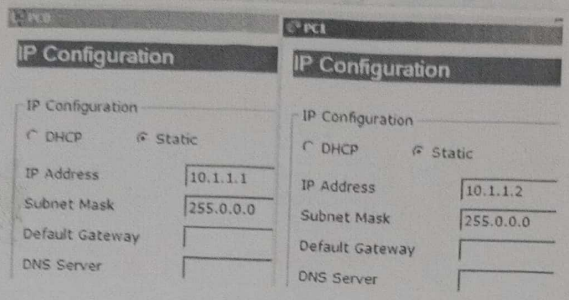
b) Select one of the PC and connect it to hub using the cable. The link LED should glow in green, indicating that the link is up. similarly connect remaining 3 PCs to the hub.

iii) Similarly connect 4 PCs to the switch using copper straight-through cable





iv) Click on the Pc's connected to hub, go to the Desktop tab, click on IP configuration and enter an IP address and subnet mask. Here the default gateway and DNS server information is not needed as there are only two end devices in the network.



v) Observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation.

vi) Repeat step 3 to step 5 for the Pc's connected to the switch.

vii) Observe how hub and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and hub.

Student observation:

a) From your observation write down the behaviour of switch and hub in terms of forwarding the packets received by them.

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⇒ Hub:

* A hub broadcasts the received packet to all ports connected to it, except the port from which the packet was received.

Switch:

* A switch checks the MAC address of the destination device and forwards the packet only to the specific port where the device is connected.

Device	Packet forwarding Behaviour.
Hub	Broadcast packets to all connected ports.
Switch	forwards packets only to the destination port.

b) Find out the network topology implemented in your college and draw and label that topology in your observation book.

⇒ Most modern college computer networks use a star topology, often extended into a hierarchical/ extended star topology.

Labels to include:

* Internet.
* Access switch
* Network cables.
* Core switch.

Observation:

The college network follows an extended star topology, where multiple computers are connected to access switches and the access switches are connected to a central switch.

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Ex.No:4 Setup and Configure a LAN using a switch and ethernet cables in your lab.

Aim:

To design, set-up and configure a local area network connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment required:

Equipment	Quantity.	specification
PC	4	Ethernet (NIC) enabled.
Network switch	1	8/16/24 port, unmanged / managed.
Ethernet cables (Cat 5e / cat 6)	4	Straight-through, RJ-45 connectors.

Theory:

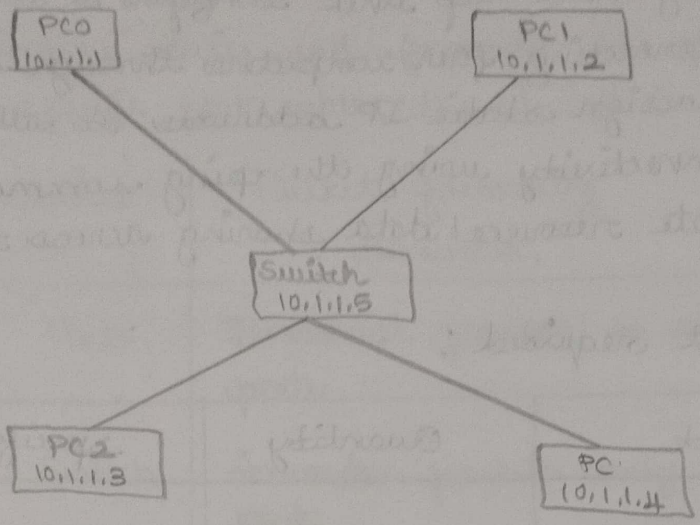
A local area network is a network that connects computers and devices within a limited geographical area, allowing them to share resources such as files, and internet connections.

A star topology is used in this experiment, where all end devices (PCs) are individually connected to a central network switch. The switch operates at layer 2 of the OSI model and forwards data frames between connected devices on the MAC addresses. Every device on a LAN must be assigned an IP address within the same network range.

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Network diagram

The LAN was implemented using a star topology as shown below, with 4 PC's connected individually to a central switch.



IP address configuration table

Device	IP address	subnet mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/1.
PC1	10.1.1.2	255.0.0.0	switch port Fa0/2.
PC2	10.1.1.3	255.0.0.0	switch port Fa0/3.
PC3	10.1.1.4	255.0.0.0	switch port Fa0/4.
switch	10.1.1.5	255.0.0.0.	VLAN 1 Interface.

Implementation in Cisco Packet Tracer:

Step 1; Place the devices.

1) Open Cisco Packet tracer.

2) From the device category panel, select End devices and drag 4 PCs onto the workspace.

3) Switch and drag 1 switch onto the workspace.

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Step 2: Cable the devices.

4. click connections → select copper straight-through cable.

5. click each PC, then click the switch, and select an available fast ethernet port to complete each connection.

6. Repeat for all four PCs. green link lights confirm a successful physical connection.

Step 3: Assign IP addresses to PCs.

7. click a PC → desktop tab → IP configuration.

8. Enter the static IP address and subnet mask as per the IP table for each PC.

Step 4: Configure the switch management IP.

9. click the switch → CLI tab.

10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using ping

11. click a PC → desktop tab → command prompt.

12. Use the ping command to test connectivity to the other PCs.

13. Successful "Reply from...." messages confirm Connectivity.

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Step 6: Demonstrate data sharing (Server simulation)

14) Add a server device (End devices → Server-PT) and connect it to the switch.

15) Assign it an IP address, subnet mask via its Desktop → IP configuration tab.

16) Open the server's services tab and enable the http service.

17) From another PC, open the web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe packet flow in simulation mode.

18) Switch to simulation mode.

19) Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC address based forwarding.

Observation:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet. Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computer on the network, confirming that the LAN was functioning correctly for resource sharing.

Ex.No:5 Experiment on Packet capture tool: Wireshark.

Aim:

To determine the packet capture tool like wireshark.

Packet Sniffer:

* Sniffs messages being sent/received from/by your computer.

* Passive program,

⇒ never sends packet itself.

⇒ no packets addressed to it.

⇒ receives a copy of all packets.

Description:

Wireshark:

* Wireshark, a network analysis tool formerly known as ethereal, captures packets in real time and display them in human readable format. Wireshark includes filters, color coding and other features that let you dig deep into network traffic and inspect individual packets.

Wireshark used for,

⇒ Network administrators: troubleshoot network problems.

⇒ Network security engineers: examine security problems.

⇒ Developers: debug protocol implementations.

⇒ People: learn network protocol internals.

Getting Wireshark:

Wireshark can be downloaded for windows or macOS from its official website. For linux or another UNIX-like system, wireshark will see found in its package repositories.

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Capturing packets:

After downloading and installing wireshark, launch it and double-click the name of a network interface under capture to start capturing packets on that interface. As soon as click the interface's name, you'll see the packets start to appear in real time.

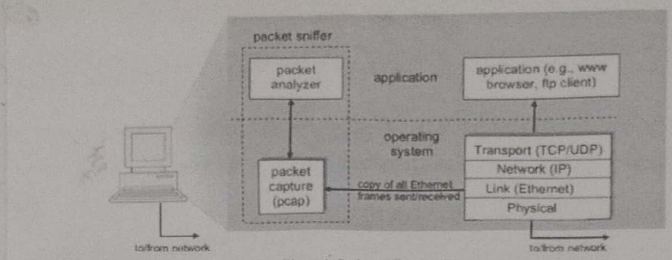
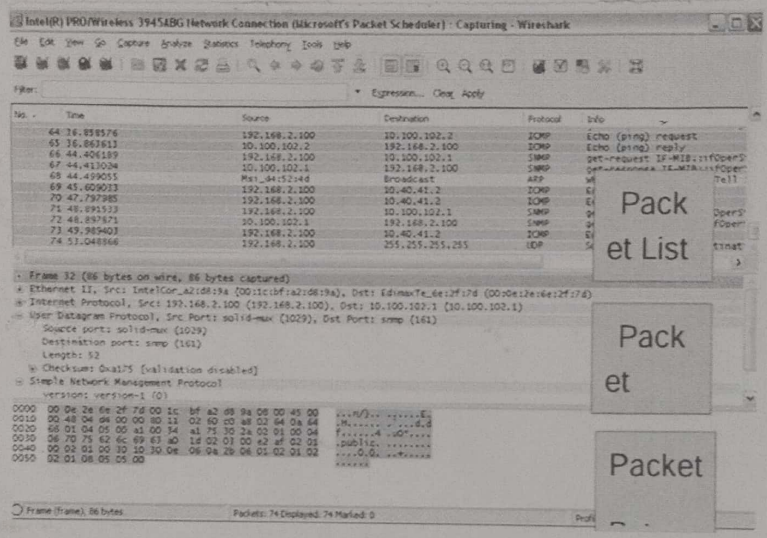


Figure 1: Packet sniffer structure



The "Packet List" Pane:

The packet list pane displays all the packets in the current capture file. The Packet list plane each line in the packet list corresponds to one packet in the capture file.

The "Packet details" pane:

The packet details pane shows the current packet in a more detailed form. The pane shows the protocol and protocol fields of the packet selected in the "Packet List" pane.

The "Packet bytes" Pane:

The packet bytes pane shows the data of the current packet in a hexdump style.

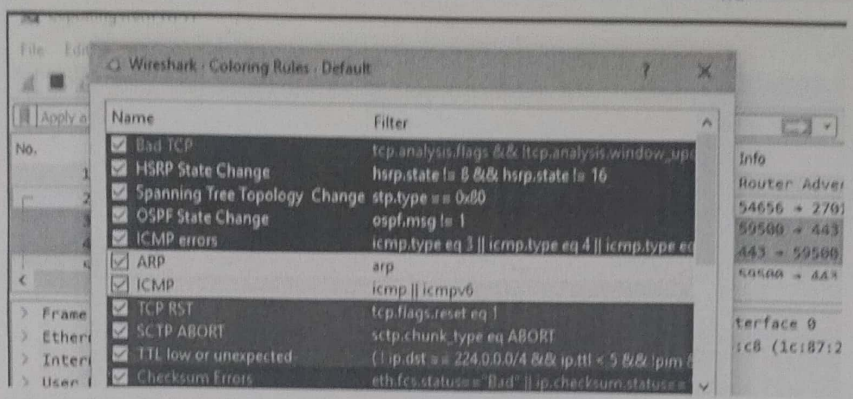
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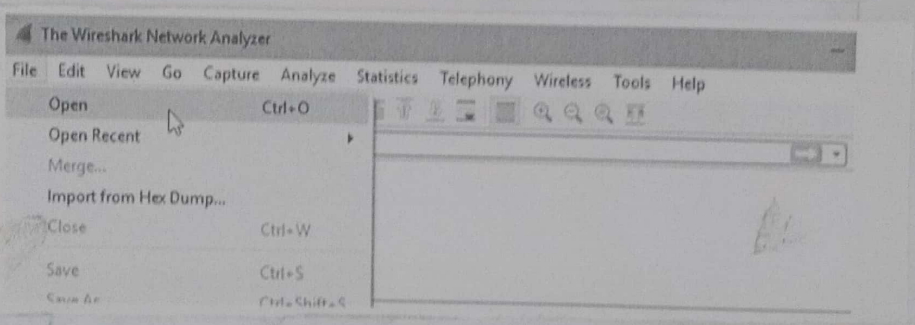
Color coding:

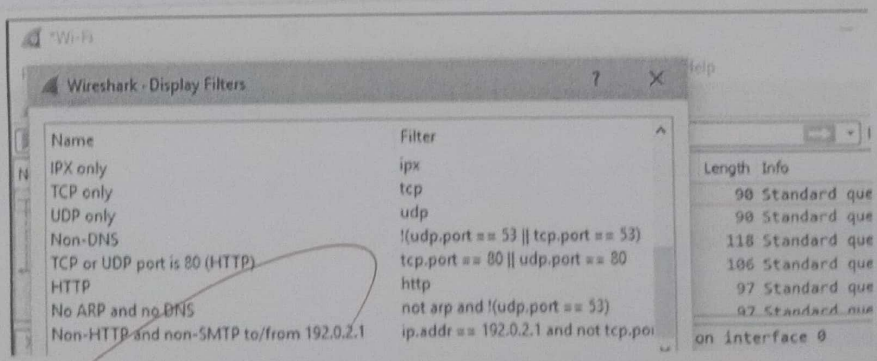
You will probably see packets highlighted in a variety of different colors. Wireshark uses colors to help you identify the types of traffic at a glance. To view exactly what the color codes mean, click View > Coloring Rules.

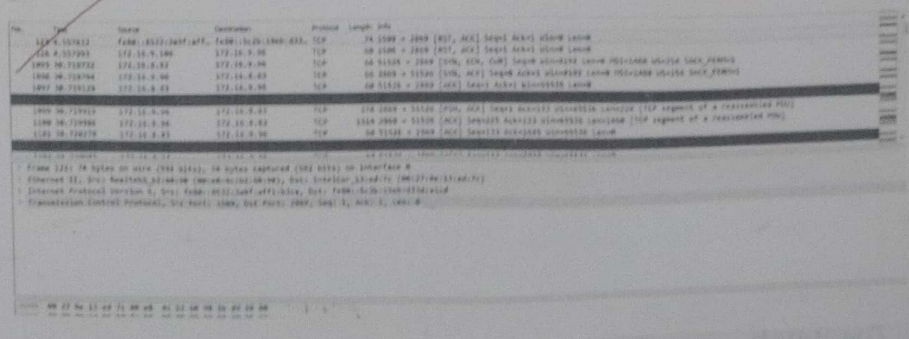
Filtering packets:

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic.









Capturing and analysing packets using wireshark tool.

To filter, capture, view, packets in wireshark tool.

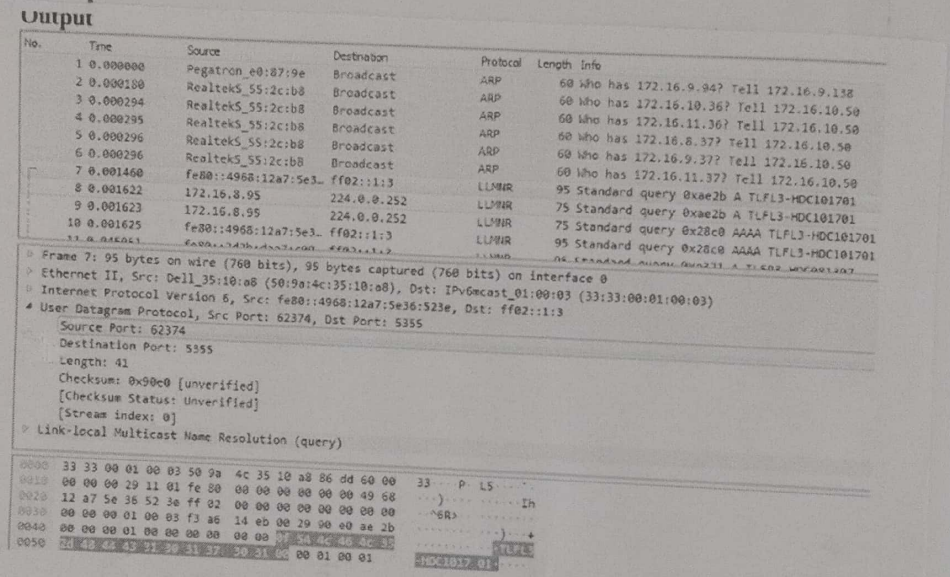
Capture 100 packets from the ethernet: IEEE 802.3 LAN interface and save it.

Procedure:

* Select local area connection in wireshark.
* Go to capture option.
* Select stop capture automatically after 100 packets
* Then click start capture.
* Save the packets.

Output:

Output



i) Create a filter to display only TCP/UDP packets, inspe the packets and provide the flow graph.

ii) Create a filter to display only ARP packets and inspect the packets.

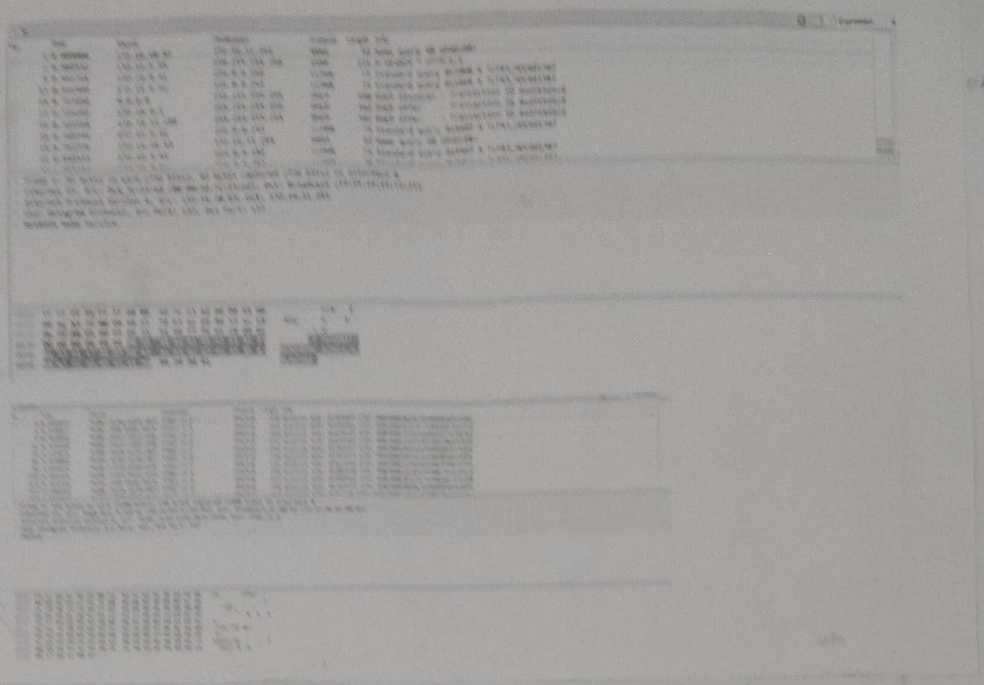
iii) Create a filter to display only DNS packets and provide the flow graph.

iv) Create a filter to display only HTTP packets and inspect the packets.

v) Create a filter to display only IP/ICMP packets and inspect the packets.

vi) Create a filter to display only DHCP packets and inspect the packets.

Scanned with OKEN Scanner



Student observation:

1) What is promiscuous mode?

⇒ It is a network interface mode in which a device captures and receives all network packets.

2) Does ARP packet have a transport layer header? Explain

⇒ No, ARP works directly at the network access and is not carried using TCP or UDP.

3) Which transport layer protocol is used by DNS?

⇒ DNS primarily uses UDP at port 53 for normal queries. It can also use TCP port 53, especially for zone transfers.

4) What is the port number used by HTTP protocol?

⇒ The port number for HTTP is 80.

5) What is a broadcast IP address?

⇒ A broadcast IP address is an IP address used to send a packet to all devices on a particular network/subnet.

Scanned with OKEN Scanner

Ex:No:6 Error correction at Data link layer.

Aim:

Write a program to implement error detection and correction using hamming code concept. Make a test run to input data stream and verify error correction.

Error correction at Data link layer:

* Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver.

Create a sender program:

⇒ Input to sender file should be a text of any length. Program should convert the text to binary.

⇒ Apply hamming code concept on the binary data and add redundant bits to it.

⇒ Save this output in a file called channel.

Create a receiver program:

⇒ Receiver program should read the input from channel file.

⇒ Apply hamming code on a binary data to check for errors.

⇒ If there is an error, display the position of the error.

⇒ Else remove the redundant bits and convert the binary data to ascii and display the Output.

Scanned with OKEN Scanner

Code : (sender).

```
#include <stdio.h>
#include <string.h>
void hanning_code (char data, char code[13])
{
  int d[9];
  int p1, p2, p4, p8;
  for (int i=0; i<8; i++){
    d[i+1] = (data >> (7-i)) & 1;
  }
  code[3] = d[1];
  code[5] = d[2];
  code[6] = d[3];
  code[7] = d[4];
  code[9] = d[5];
  code[10] = d[6];
  code[11] = d[7];
  code[12] = d[8];
  p1 = code[3] ^ code[5] ^ code[7] ^ code[9] ^ code[11];
  p2 = code[3] ^ code[6] ^ code[7] ^ code[10] ^ code[11];
  p4 = code[5] ^ code[6] ^ code[7] ^ code[12];
  p8 = code[9] ^ code[10] ^ code[11] ^ code[12];
}
int main(){
  file *fp;
  char text[1000];
  char code[13];
  printf("Enter the text: ");
  fgets(text, sizeof(text), stdin);
  fp = fopen("channel.txt", "w");
  if (fp == NULL)
  {
    printf("Error opening channel");
    return 1;
  }
```